

PRODUCT YIELDS (DESIGN CASE)

Product	kBPSD	wt%	t/h	API
LPG	1.7	1.1	6.5	120.8
Light naphtha	5.9	4.9	27.6	69.3
Heavy naphtha	16.3	14.2	80.8	57.0
Kerosene	13.0	12.1	68.6	46.2
Diesel	14.7	14.4	81.6	37.2
AGO	7.9	8.1	45.8	30.5
LVGO	8.5	8.9	50.4	25.9
HVGO	14.6	15.9	90.4	19.3
Slop wax	1.2	1.4	7.9	12.3
Vacuum residue	16.2	19.1	108.3	8.5

CRUDE TANKAGE

(OSBL)
Arab Light 33.4 API

DESALTING & PREHEAT

(AREA 100)
Cold train E-101...105
2-stage desalter D-101A/B
Hot train E-106...111
CIT 274 C

ATM. HEATER

H-101
60.7 MW abs.
COT 366 C

ATMOSPHERIC

FRACTIONATION
C-101 + strippers
C-102/103/104
Top 1.2 barg
FZ 361 C

NAPHTHA

STABILISATION &
SPLITTING
C-105 Stabiliser
C-106 Splitter

UTILITIES

(OSBL supply)
HP/MP/LP steam, BFW
Cooling water 32/43 C
Fuel gas, N2, inst. air
Power 13.8 kV, 60 Hz
Flare / SWS / WWT

NOTES

- Flows are design case (Arab Light, 100,000 BPSD), normal operation.
- kBPSD = thousand barrels per stream day at 15 C. t/h = tonne/hour.
- Stream numbers in diamonds refer to H&MB CFU-000-PR-HMB-001.
- Dashed blocks/lines are outside battery limits (OSBL).

REV	DATE	DESCRIPTION	
A	2026-10-02	Issued for review (FEED)	
GENERIC REFINERY - GULF COAST 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT			
BLOCK FLOW DIAGRAM CRUDE & VACUUM DISTILLATION UNIT - OVERALL			
DISCIPLINE	STAGE	SCALE	SHEET
PROCESS	FEED	NTS	1 OF 1
DRAWN	CHECKED	APPROVED	DATE
CLAUDE CODE	-	-	2026-10-02
DRAWING NUMBER			REV
CFU-000-PR-BFD-001			A